

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF PHYSICS

MIDSESSION EXAMINATION

PHYS3114 – Electrodynamics

Term 3, 2022

1. Time allowed – 50 minutes {+ 15 min for download/reading + 30 min for capture/upload}
2. Total number of questions – 2
3. Total marks available – 30 (pro rata to 15% of course assessment)
4. You are allowed to use your textbook and notes during the exam.
5. Prepare a single file (pdf files preferred) with all of your responses to upload to Moodle by the end of the allotted exam time.
6. Additional time will be allocated for submitting your answers
7. Your submitted file should have your name and zID on the first page of your submitted file (preferably in the header or footer on every page).
8. When uploading please use the following naming convention: PHYS3114 zID SURNAME NAME (e.g., PHYS3114 z5556789 Riddle Tom). NOTE: failure to name your exam file correctly may result in marking delays if your file is misplaced.
9. **Candidates must verify the upload is successful and contact the exam convener immediately if there is an issue.**
10. Answer all questions concisely and legibly.
11. In some cases, sketching your answer can be helpful.
12. Questions are not all worth the same marks; questions are worth the marks shown.
13. You may use a university-approved calculator.

PHYS3114 — Definitions and Formulae

$$F(s) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x s} dx \quad (1)$$

$$\mathcal{F}\{f(x)\} = 2 \int_0^\infty f_e(x) \cos(2\pi s x) dx - 2i \int_0^\infty f_o(x) \sin(2\pi s x) dx \quad (2)$$

$$V(x', y') = C_1 C_2 \lambda^2 f_o^2 \iint_{\text{bfp}} F\left(\frac{\zeta}{\lambda f_o}, \frac{\eta}{\lambda f_o}\right) e^{-i2\pi(\frac{f_g}{D} x' \frac{\zeta}{\lambda f_o} + \frac{f_o}{D} y' \frac{\eta}{\lambda f_o})} d\left(\frac{\zeta}{\lambda f_o}\right) d\left(\frac{\eta}{\lambda f_o}\right) \quad (3)$$

$$\gamma(P_1, P_2) = \frac{\iint_{\text{source}} I(\zeta, \eta) e^{\frac{-2\pi i}{\lambda}(p\zeta + q\eta)} d\zeta d\eta}{\iint_{\text{source}} I(\zeta, \eta) d\zeta d\eta} \quad (4)$$

$$F(\rho) = 2\pi \int_0^\infty f(r) J_0(2\pi \rho r) r dr \quad (5)$$

$$\frac{m\lambda}{2} = \frac{r_m^2}{2} \left(\frac{1}{r_0} + \frac{1}{b_0} \right) \quad (6)$$

$$\frac{d^2 P}{dt^2} + \gamma \frac{dP}{dt} + \omega_0^2 P = \frac{e^2 N}{m} \left(E + \frac{\nu P}{\epsilon_0} \right). \quad (7)$$

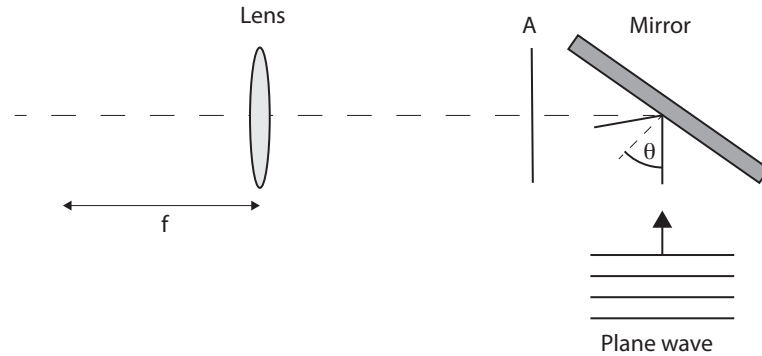
$$\hat{K} - 1 = \frac{\omega_p^2(\omega_{0'}^2 - \omega^2)}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} + i \frac{\omega_p^2 \omega \gamma}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (8)$$

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a }F\left(\frac{s}{a}\right)$
Linearity	$\alpha \cdot f(x) + \beta \cdot g(x)$	$\alpha \cdot F(s) + \beta \cdot G(s)$
Shift	$f(x - a)$	$e^{-2\pi ias}F(s)$
Convolution	$f(x) * g(x)$	$F(s)G(s)$
Derivative	$\frac{d}{dx}(f(x))$	$2\pi isF(s)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx$	$\int_{-\infty}^{+\infty} F(s) ^2 ds$

Table 1: Basic theorems of Fourier theory

Question 1 (15 marks)

An optical arrangement consists of a flat mirror reflecting an incident plane wave towards a thin lens of focal length, f . The lens has a diameter of D and the light is monochromatic with a wavelength, λ . When the wave is incident on the mirror at θ_0 (with respect to the surface normal) the reflected wave passes along the optical axis of the lens. For the following problems, the mirror is tilted such that the reflected field is deflected by a small angle with respect to the optical axis of the lens.

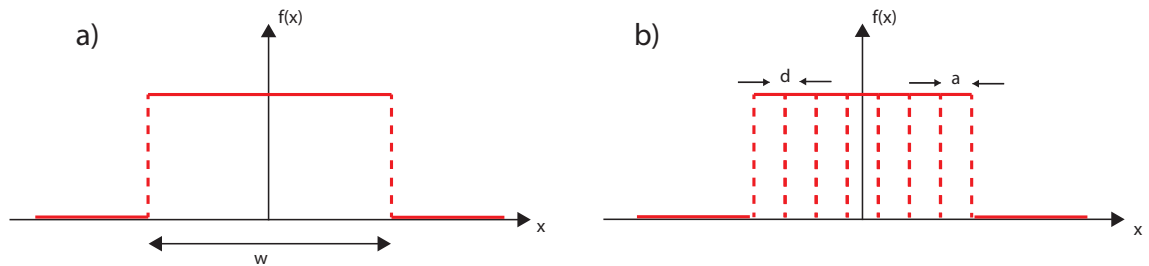


- (a) Describe qualitatively what you expect will happen to the field in the back focal plane of the lens when the mirror is tilted away from θ_0 .
- (b) Use the matrix method from geometric optics to derive an expression for the change in the focus as a function of mirror tilt angle.
- (c) By considering the field at plane A (or otherwise), use Fourier diffraction theory to derive an expression for the diffraction pattern that is observed at the back focal plane as a function of mirror tilt angle.
- (d) Compare the results from the Geometric and Fourier optics methods and comment on any similarities and differences.

Question 2 (15 marks)

Aperture functions used in scalar diffraction problems can be constructed in a number of ways using a combination of Fourier theorems and transform pairs. However, regardless of the mathematical representation, they should yield the same diffraction pattern.

- (a) Show that a displaced double delta function $f(x) = \delta(x - b) + \delta(x - c)$ can be represented by a linear combination of odd and even functions.
- (b) Using this combination of functions, calculate the corresponding Fourier transform and show that it is equivalent to the application of the Shift Theorem to a double delta function centred about the origin.
- (c) A single open slit can be represented as either a single hat function or as a finite array of smaller hat functions (see Figure).



What relationship between the width of the original slit, w , and the number, n , spacing, d , and width, a , of the array of hats would make the two functions equivalent?

- (d) By calculating the Fourier transform of the array of hat functions, show that it is equivalent to the transform of a single hat function.